Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

IZOVAT Thermal insulation Light, 25-45 kg/m^3

From OBIO LLC

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
Type of EPD	EPD of a single product from a manufacturer
EPD registration number:	EPD-IES-0027479
Publication date:	2025-12-09
Valid until:	2030-12-08
	An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com



GENERAL INFORMATION

Programme Information	
Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	support@environdec.com

Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): ***Construction Products, PCR 2019:14 Version 2.0.1***
PCR review was conducted by: ***The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción (Chile). The review panel may be contacted via the Secretariat info@environdec.com***
c-PCR, if applicable: ***c-PCR-005 (TO PCR 2019:14) Thermal Insulation Products (EN 16783:2024)***

Third-party Verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ **Individual EPD verification without a pre-verified LCA/EPD tool** Third-party verifier: *Elisabet Amat, GREENIZE* Approved by: International EPD System
Procedure for follow-up of data during EPD validity involves third-party verifier: ☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.
For further information about comparability, see EN 15804 and ISO 14025.

INFORMATION ABOUT EPD OWNER

Owner of the EPD: LLC OBIO (IZOVAT)

Address:
Legal address: Promyslova 6, 10025, Zhytomyr, Ukraine
Manufacturing site: Promyslova 6, 10025, Zhytomyr, Ukraine

Contact: Mariia Zhydkova, +38 (095) 283-11-88, quality@izovat.ua

Address and contact information of the LCA practitioner commissioned by the EPD owner:
Bureau Veritas Latvia SIA. Email: riga@bureauveritas.com

Description of the organization: OBIO LLC has been operating in the basalt thermal insulation market for over 15 years, establishing itself as a reliable partner and professional in its field. The company has implemented modern management methods in accordance with international standards, ensuring systematic improvement of its operations and enhancing competitiveness. OBIO LLC continually works to improve product quality and reduce the environmental impact of its production, with a particular focus on preventing and minimizing pollution. The primary objective of the company is to produce and sell high-quality, competitive products under the IZOVAT trademark, in accordance with applicable standards and certification requirements.

Product-related or management system-related certifications: All IZOVAT™ products are certified in accordance with the requirements of EN 13162:2012 + A1:2015. The company's quality management system is certified in accordance with the international standard EN ISO 9001:2015 by the certification bodies CQS (Certified Quality Systems) and IQNet (International Certification Network).

PRODUCT INFORMATION

Product name: IZOVAT™ Thermal Insulation Light, 25-45 kg/m^3

Product identification: Each product, during its manufacture, is assigned a batch number, date of manufacture and a unique identification code in accordance with the requirements of the European standard EN 13162:2012+A1:2015.

Visual representation (e.g., an image) of the product



UN CPC code: 37990 - Non-metallic mineral products n.e.c. (including mineral wool, expanded mineral materials, worked mica, articles of mica, non-electrical articles of graphite or other carbon, and articles of peat).

Product Description:
IZOVAT™ stone wool thermal insulation is a fire-safe material used for protection against heat, cold, fire, vibrations, and noise. It is traditionally manufactured from volcanic rock (typically basalt or dolomite) with a low percentage of resin binder. The stone wool thermal insulation materials described in this declaration are produced in the form of slabs with densities ranging from 25 to 45 kg/m³. Packaging materials and additives are included in this declaration. IZOVAT™ Light is suitable for thermal and soundproofing insulation of internal building components, such as attics, pitched roofs, and frame partitions, where the thermal insulation material is not exposed to external influences. Installed between structural or framing elements.

Classification of reaction to fire of OBIO LLC products - A1

Name and location of production site(s): Promyslova 6, 10025, Zhytomyr, Ukraine

CONTENT DECLARATION

Product content	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic material, mass-% of product	Biogenic material, kg C/product or functional unit
Raw materials (Basalt, dolomite, slag)	4.01	0%	0%	-
Binders and phenolic resins	0.13	0%	0%	-
Additives	0.1	0%	0%	-
TOTAL	4.2	0%	0%	-

Packaging materials	Mass, kg	Mass-% (versus the product)	Biogenic material, kg C/product or functional unit
Wood pallets	0.47	11.10%	0.23
Carton board	0.0046	0.11%	0.002
Plastic film	0.11	2.60%	-
TOTAL	0.58	13.80%	0.24

1 kg of biogenic carbon in the product/packaging is equivalent to the uptake of 44/12 kg of CO_2.

The product does not contain any hazardous substance listed on the "Candidate List of Substances of Very High Concern (SVHC) for Authorization" in amounts greater than 0.1% (1000 ppm).

LCA INFORMATION

Declared unit: 1 m^2 of insulation slab with an average density of 35 kg/m^3, an average thickness of 121 mm, and a thermal resistance of 2.30 $m^2/K/W$, considering the declared thermal conductivity index of 0.037W/mK. Weight – 4.24 kg/m^2, conversion factor to mass – 0.2358

Reference service life: 50 years

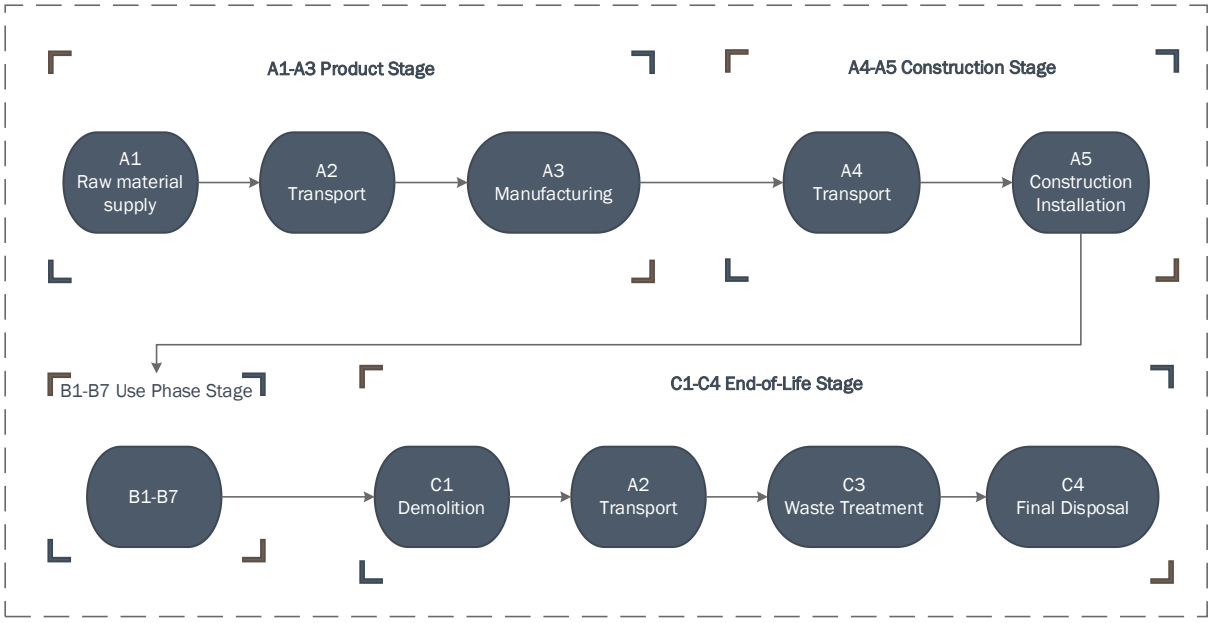
Time representativeness: The activity data for the life cycle inventory representing the processes in modules A1-A3 correspond to the reporting period for the calendar year 2023.

Geographical scope: The activities modelled in this EPD correspond to Ukraine for modules A1-A3 and a European scenario for all other stages in the life cycle (A4-A5, B, C, D)

Database(s) and LCA software used: The LCA for the EPD development was performed in SimaPro 9.6 with Ecoinvent 3.10 and the CF package for EN 15804 based on the EF 3.1.

Description of system boundaries:
c) Cradle to Grave and module D (A+B+C+D). The scenarios included are currently in use and are representative of one of the most likely scenario alternatives.

Process flow diagram:



More information: During this LCA, the *polluters pay,* and modularity principles have been followed. Double counting has been avoided. This EPD is intended for a specific product rather than for multiple products.

The processes related to infrastructure, construction, and production of equipment and tools that are not directly consumed in the production process have been excluded. Activities related to personnel, such as transportation to and from work, have been excluded.

Data quality: The foreground data was collected internally, considering the available production amounts and measures during the reporting period.

According to the UN Environment Global Guidance criteria on LCA database development, the quality level is good. The data is geographically representative, as it originates from the area of study. It is technically representative as it comes from processes and products under study using the same state of technology defined in the goal and scope. According to the documentation, it is also time-representative as the time distance between the valid data from the background and the reporting period is no more than three years. A data quality rating was performed using a rating system where one means very good and five means very poor, in line with EN 15804 Annex E and the guidelines for EF-compliant datasets and quality data assessment. An average for each criterion is presented as follows:

Technological Representativeness, TeR	Geographic representativeness, GeR	Time Representativeness, TiR	Precision, P	Average DQR
2.71	2.23	2.32	2.06	2.33

Data Quality Declaration as per the PCR 2019:14[1], section 4.6.5, and EN 15491:

Process	Source Type	Source	Reference year	Data Category	Share of primary data, of GWP-GHG results for A1-A3
Direct emissions from manufacturing	Collected data	EPD Owner	2023	Primary data	47.7 %
Foundry coke	EPD	EPD Reg. Number: 3015-EPD-030066839	2024	Primary data	0.3 %
Waste treatment from manufacturing	Database	Ecoinvent 3.10	2024	Secondary data	0%
Total share of primary data, of GWP-GHG results for A1-A3					48.1 %

Cut-off criteria: All primary raw materials and processes have been considered. A cut-off has been applied in this study, excluding labels and ink for packaging. The materials subject to cut-off do not contain dangerous substances and correspond to less than 0.1 % of the total mass per declared unit.

Allocation: EN 15804 and PCR 2019:14 recommend that allocation among products and co-products should be avoided when possible. However, given the nature of the manufacturing process, some flows, such as ancillary and packaging materials, electricity, and production of output flows to waste treatment activities, were allocated based on physical properties (production volumes by mass).

Stages and Production Description

Manufacturing Summary: Stone wool, also known as mineral wool, is manufactured by melting volcanic rock and other raw materials at high temperatures, then spinning the molten mixture into fine fibers. These fibers are then collected, compressed, and formed into various shapes such as rolls, batts, or boards. This process is inspired by volcanic eruptions that create strands of volcanic glass.
The manufacturing process includes the following steps:

1. Raw Material Preparation: Basalt rock, dolomite and slag from the steel industry are combined and proportioned.
2. Melting: The raw materials are melted in a furnace at extremely high temperatures, typically between 1300°C and 1500°C.

[1] The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

3. Fiber Formation (Spinning): The molten material is spun into fibers using rotating flywheels or spinners—a process similar to making cotton candy, but with molten rock.
4. Cooling and Curing: The fibers are cooled, typically with air or water, and then hardened in a curing oven at about 200°C.
5. Cutting and Shaping: The resulting material is cut into the desired shapes and sizes (rolls, batts, boards).
6. Packaging and Shipping: The finished products are packaged and prepared for shipping. The final product is a lightweight, fire-resistant, and sound-absorbing insulation material.

Product Stage

A1 – Raw material supply: This stage considers the extraction and processing of all raw materials, including secondary material processing, if applicable.

A2 - Transport: This stage accounts for transporting raw materials to the facility in Ukraine. It includes road transport by lorry and freight train vehicles.

A3 - Manufacturing: This stage includes the manufacturing process conducted in the facilities before transporting the finished product to different locations. The input of packaging materials is considered in this stage. Waste treatment activities derived from manufacturing waste are also accounted for until such elements reach the end of the waste state.

The electricity generation and upstream processes linked to extracting and transporting materials and fuels prior to combustion or utilization are automatically considered within the Ecoinvent dataset. The electricity used at the manufacturing module (A3) has been modeled following a market-based approach and in accordance with the contractual instrument in place, reflecting a residual mix. The electricity mix representing the background dataset used includes the transmission network, direct emissions to air of Ozone, and N_2O. The modeled electricity mix is as follows:

Electricity Source	Share in the electricity mix
Coal	30.67 %
Natural Gas	0.62 %
Fuel Oil	0.00 %
Nuclear	40.61 %
Other carbon-free sources	0.89 %
Other Renewable energy sources	27.21 %

The resulting impact from the modeled customized electricity mix, utilizing the GWP-GHG indicator, is 0.529 kg CO_2eq/kWh.

Construction Stage

A4 – Transport to the construction site: This module includes the transport to the different countries where the product was purchased during the reporting period (2023). The detailed transport table is presented as follows:

Product	Distance 1 (km)	Transport type	Destination
IZOVAT Light 25-45 kg/m^3	730	truck	Romania
	510	truck	Moldova
	280	truck	Ukraine

A5 – Construction/Installation: This module is included in the LCA model, accounting for the packaging materials' EoL activities. The final fate of packaging materials accompanying the main product is assessed in accordance with PCR 2019:14 V2.0.1, Annex 2, to balance out the biogenic carbon content in the packaging materials. Transport to the waste treatment facility/landfill and the energy leaving the product system are accounted for, as recommended in Annex 3 of PCR 2019:14 V 2.0.1, following option B. It has been assumed that 2 % of the main product is lost during installation.

In this module, it is assumed that 50% of the biogenic packaging materials (wood pallets and carton corners) are sent to incineration with an efficiency inferior to 60%. The remaining 50 % of biogenic packaging materials are modeled according to the average European waste treatment scenario as per Ecoinvent.

Plastic packaging materials are modeled according to the average European waste treatment scenario as per Ecoinvent.

Use Stage (B1-B7):

During a regular-use scenario, it is assumed that no maintenance, repair, replacement, and/or refurbishment is required. Neither energy or water use is required in modules B6 and B7.

End of Life Stage:

C1 – Demolition/Deconstruction: This module examines the energy requirements and potential emissions associated with demolition activities related to the product under study. For this particular case, secondary representative background data was used based on the Ecoinvent process for demolition of “mineral wool, recycling Europe without Switzerland”. Additionally, the energy required for demolition was derived from the proxy value reported in Table 4 of PCR 2019:14 V2.0.1.

C2 – Transport: Transport to the waste disposal treatment site. A distance of 80 km, as per Table 4 in PCR 2019:14, was assumed in a EURO 5 truck.

C3 – waste processing: This module does not consider any activity associated with waste treatment for further use or recycling.

C4 – disposal: This module accounts for the final disposal of the IZOVAT insulation materials to reach the end-of-waste state. It is assumed that the IZOVAT™ product is transported for sorting and final disposition in a landfill. In this module, the electricity consumption for sorting, the sorting facility, and the handling in the sorting plant are considered. Additionally, this module accounts for the transportation from the sorting facility to the landfill (77 km assumed) and the landfilling of inert material.

Benefits and loads beyond the system boundaries:

Module D declares the environmental benefits of reusable products, recyclable materials, or energy recovery. This module accounts for the benefits related to the hypothetical use and production avoidance of energy exported from the incineration with energy recovery of packaging materials (wood pallets) in module A5.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Geography	GLO	GLO	UA	EU	EU	EU	EU	EU	EU	EU	EU	EU	EU	EU	EU	EU	EU
Variation - products	0 %			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - sites	0 %			-	-	-	-	-	-	-	-	-	-	-	-	-	-

Description of the system boundary (X = Included in LCA; ND = Not declared; NR = Not relevant)

Results of the Environmental Performance Indicators

The following results represent the potential environmental impact of the IZOVAT™ Light Thermal Insulation product, considering the system boundaries, limitations, and assumptions in this EPD and its LCA project report.

Mandatory impact category indicators according to EN 15804+A2

Results per functional or declared unit								
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**
GWP-Total	kg CO_2 eq.	3.12E+00	2.86E-01	4.64E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
GWP-fossil	kg CO_2 eq.	2.91E+00	2.86E-01	1.53E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
GWP-biogenic	kg CO_2 eq.	2.13E-01	5.11E-05	3.11E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
GWP-luluc	kg CO_2 eq.	8.50E-04	9.33E-05	1.89E-05	0.0E+00	0.0E+00	0.0E+00	0.0E+00
ODP	kg CFC 11 eq.	5.25E-08	5.68E-09	6.38E-10	0.0E+00	0.0E+00	0.0E+00	0.0E+00
AP	mol H^+ eq.	6.00E-03	8.93E-04	2.14E-04	0.0E+00	0.0E+00	0.0E+00	0.0E+00
EP-freshwater	kg P eq.	2.32E-04	1.91E-05	5.78E-06	0.0E+00	0.0E+00	0.0E+00	0.0E+00
EP-marine	kg N eq.	4.06E-03	3.01E-04	8.98E-05	0.0E+00	0.0E+00	0.0E+00	0.0E+00
EP-terrestrial	mol N eq.	1.64E-02	3.27E-03	8.18E-04	0.0E+00	0.0E+00	0.0E+00	0.0E+00
POCP	kg NMVOC eq.	1.23E-02	1.40E-03	2.94E-04	0.0E+00	0.0E+00	0.0E+00	0.0E+00
ADP-minerals& metals*	kg Sb eq.	4.29E-06	9.13E-07	2.93E-07	0.0E+00	0.0E+00	0.0E+00	0.0E+00
ADP-fossil*	MJ	3.66E+01	4.01E+00	5.58E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
WDP*	m^3	4.02E-01	1.64E-02	1.19E-03	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Acronyms	**GWP-fossil** = Global Warming Potential fossil fuels; **GWP-biogenic** = Global Warming Potential biogenic; **GWP-luluc** = Global Warming Potential land use and land use change; **ODP** = Depletion potential of the stratospheric ozone layer; **AP** = Acidification potential, Accumulated Exceedance; **EP-freshwater** = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; **EP-marine** = Eutrophication potential, fraction of nutrients reaching marine end compartment; **EP-terrestrial** = Eutrophication potential, Accumulated Exceedance; **POCP** = Formation potential of tropospheric ozone; **ADP-minerals&metals** = Abiotic depletion potential for non-fossil resources; **ADP-fossil** = Abiotic depletion for fossil resources potential; **WDP** = Water (user) deprivation potential, deprivation-weighted water consumption							

Indicator	Unit	B5	B6	B7	C1	C2	C3	C4	D
GWP-Total	kg CO_2 eq.	0.0E+00	0.0E+00	0.0E+00	1.68E-03	6.57E-02	0.00E+00	1.36E-01	-2.62E-02
GWP-fossil	kg CO_2 eq.	0.0E+00	0.0E+00	0.0E+00	1.68E-03	6.57E-02	0.00E+00	1.36E-01	-2.62E-02
GWP-biogenic	kg CO_2 eq.	0.0E+00	0.0E+00	0.0E+00	1.50E-07	1.18E-05	0.00E+00	2.97E-05	-5.63E-06
GWP-luluc	kg CO_2 eq.	0.0E+00	0.0E+00	0.0E+00	1.46E-07	2.15E-05	0.00E+00	4.59E-05	-2.91E-06
ODP	kg CFC 11 eq.	0.0E+00	0.0E+00	0.0E+00	2.57E-11	1.31E-09	0.00E+00	2.88E-09	-4.06E-10
AP	mol H^+ eq.	0.0E+00	0.0E+00	0.0E+00	1.52E-05	2.05E-04	0.00E+00	9.13E-04	-7.09E-05
EP-freshwater	kg P eq.	0.0E+00	0.0E+00	0.0E+00	4.91E-08	4.38E-06	0.00E+00	1.10E-05	-9.43E-06
EP-marine	kg N eq.	0.0E+00	0.0E+00	0.0E+00	7.04E-06	6.92E-05	0.00E+00	3.85E-04	-1.74E-05
EP-terrestrial	mol N eq.	0.0E+00	0.0E+00	0.0E+00	7.71E-05	7.53E-04	0.00E+00	4.20E-03	-1.74E-04
POCP	kg NMVOC eq.	0.0E+00	0.0E+00	0.0E+00	2.30E-05	3.22E-04	0.00E+00	1.37E-03	-5.30E-05
ADP-minerals &metals*	kg Sb eq.	0.0E+00	0.0E+00	0.0E+00	6.00E-10	2.10E-07	0.00E+00	2.65E-07	-7.90E-09
ADP-fossil*	MJ	0.0E+00	0.0E+00	0.0E+00	2.20E-02	9.22E-01	0.00E+00	2.16E+00	-4.13E-01
WDP*	m^3	0.0E+00	0.0E+00	0.0E+00	4.77E-05	3.78E-03	0.00E+00	3.43E-02	-2.13E-03
Acronyms	**GWP-fossil** = Global Warming Potential fossil fuels; **GWP-biogenic** = Global Warming Potential biogenic; **GWP-luluc** = Global Warming Potential land use and land use change; **ODP** = Depletion potential of the stratospheric ozone layer; **AP** = Acidification potential, Accumulated Exceedance; **EP-freshwater** = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; **EP-marine** = Eutrophication potential, fraction of nutrients reaching marine end compartment; **EP-terrestrial** = Eutrophication potential, Accumulated Exceedance; **POCP** = Formation potential of tropospheric ozone; **ADP-minerals&metals** = Abiotic depletion potential for non-fossil resources; **ADP-fossil** = Abiotic depletion for fossil resources potential; **WDP** = Water (user) deprivation potential, deprivation-weighted water consumption								

** Disclaimer. The results of modules A1-A3 should not be used without considering the results of modules C1-C4.*
*** The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or there is limited experience with the indicator.*
**** The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.*

Additional mandatory and voluntary impact category indicators

Results per functional or declared unit								
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**
GWP-GHG[2]	kg CO_2 eq.	2.91E+00	2.86E-01	1.53E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00

Results per functional or declared unit									
Indicator	**Unit**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG	kg CO_2 eq.	0.0E+0	0.0E+0	0.0E+0	1.68E-03	6.57E-02	0.00E+00	1.36E-01	-2.62E-02

Resource use indicators

Results per functional or declared unit								
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**
PERE	MJ	5.47E-01	5.27E-02	1.21E-02	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PERM	MJ	2.68E+00	0.00E+00	-4.34E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PERT	MJ	3.23E+00	5.27E-02	-4.22E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PENRE	MJ	3.92E+01	4.01E+00	5.58E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PENRM	MJ	2.35E+00	1.48E-04	1.83E-05	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PENRT	MJ	4.16E+01	4.01E+00	5.58E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
FW	m^3	1.65E-02	5.49E-04	1.78E-04	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Acronyms	**PERE** = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; **PERM** = Use of renewable primary energy resources used as raw materials; **PERT** = Total use of renewable primary energy resources; **PENRE** = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; **PENRM** = Use of non-renewable primary energy resources used as raw materials; **PENRT** = Total use of non-renewable primary energy re-sources; **SM** = Use of secondary material; **RSF** = Use of renewable secondary fuels; **NRSF** = Use of non-renewable secondary fuels; **FW** = Use of net fresh water							

[2] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Results per functional or declared unit									
Indicator	**Unit**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	0.0E+0	0.0E+0	0.0E+0	1.09E-04	1.21E-02	0.00E+00	3.73E-02	-7.70E-03
PERM	MJ	0.0E+0	0.0E+0	0.0E+0	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	0.0E+0	0.0E+0	0.0E+0	1.09E-04	1.21E-02	0.00E+00	3.73E-02	-7.70E-03
PENRE	MJ	0.0E+0	0.0E+0	0.0E+0	2.20E-02	9.22E-01	0.00E+00	2.16E+00	-4.13E-01
PENRM	MJ	0.0E+0	0.0E+0	0.0E+0	1.54E-07	3.41E-05	0.00E+00	8.34E-05	-2.95E-06
PENRT	MJ	0.0E+0	0.0E+0	0.0E+0	2.20E-02	9.22E-01	0.00E+00	2.16E+00	-4.13E-01
SM	kg	0.0E+0	0.0E+0	0.0E+0	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.0E+0	0.0E+0	0.0E+0	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.0E+0	0.0E+0	0.0E+0	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	0.0E+0	0.0E+0	0.0E+0	1.57E-06	1.26E-04	0.00E+00	9.05E-04	-3.17E-04
Acronyms	**PERE** = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; **PERM** = Use of renewable primary energy resources used as raw materials; **PERT** = Total use of renewable primary energy resources; **PENRE** = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; **PENRM** = Use of non-renewable primary energy resources used as raw materials; **PENRT** = Total use of non-renewable primary energy resources; **SM** = Use of secondary material; **RSF** = Use of renewable secondary fuels; **NRSF** = Use of non-renewable secondary fuels; **FW** = Use of net fresh water								

Waste indicators

Results per functional or declared unit								
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**
Hazardous waste disposed	kg	1.93E-04	2.70E-05	1.18E-05	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Non-hazardous waste disposed	kg	5.76E+01	1.90E-01	1.01E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Radioactive waste disposed	kg	1.56E-05	1.28E-06	2.52E-07	0.0E+00	0.0E+00	0.0E+00	0.0E+00

Results per functional or declared unit									
Indicator	**Unit**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	0.0E+0	0.0E+0	0.0E+0	1.52E-07	6.21E-06	0.00E+00	1.39E-05	-6.96E-07
Non-hazardous waste disposed	kg	0.0E+0	0.0E+0	0.0E+0	1.35E-05	4.38E-02	0.00E+00	4.28E+00	-6.15E-04
Radioactive waste disposed	kg	0.0E+0	0.0E+0	0.0E+0	2.42E-09	2.94E-07	0.00E+00	9.87E-07	-1.44E-06

Output flow indicators

Results per functional or declared unit								
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**
Components for re-use	kg	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Material for recycling	kg	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Materials for energy recovery	kg	0.0E+0	0.0E+0	2.4E-01	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Exported energy, electricity	MJ	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Exported energy, thermal	MJ	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0

Results per functional or declared unit									
Indicator	**Unit**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Material for recycling	kg	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Materials for energy recovery	kg	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Exported energy, electricity	MJ	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0
Exported energy, thermal	MJ	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0	0.0E+0

Abbreviations

Abbreviation	Definition
General Abbreviations	
EN	European Norm (Standard)
EF	Environmental Footprint
GPI	General Programme Instructions
ISO	International Organization for Standardization
CEN	European Committee for Standardization
CPC	Central product classification
SVHC	Substances of Very High Concern
ND	Not Declared
EPD	Environmental Product Declaration
PCR	Product Category Rules
c-PCR	Complementary Product Category Rules
UN	United Nations
GHG	Greenhouse Gases
EoL	End-of-Life
EU	Europe
GLO	Global
UA	Ukraine

References

- General Programme Instructions of the International EPD® System. Version 5.0.
- PCR 2019:14. Construction Products. Version 2.0.1
- c-PCR 05 Thermal Insulation Products (EN 16783:2024)
- ISO 14040:206: Environmental Management-Life Cycle Assessment-Principles and framework.
- ISO 14044:206/Amd 2:2020 Environmental management. Life Cycle Assessment. Requirements and guidelines
- ISO 14025:206 Environmental labels and declarations. — Type III environmental declarations — Principles and procedures
- EN 15804:2012+A2:2019/AC:2021 – Sustainability of construction Works – Environmental product declarations – Core rules for the product category of construction products.
- Ecoinvent v3.10 Database.
- Fazio, Simone, et al. Guide for EF-compliant data sets (Version 2.0). Publications Office of the European Union, 2020. doi:10.2760/537292.
- LCA Project Report –MINERAL WOOL IZOVAT™ Thermal Insulation products by OBIO LLC

Version History

Original Version of the EPD, 2020-05-11
MRPI® registration
1.1.00122.2020

Revision 1, 2025-12-09
Differences from the previously published version: The reporting period has been updated to calendar year 2023. Foreground and background data were updated to represent the new reporting period.

EPD
INTERNATIONAL EPD SYSTEM

www.environdec.com